



Fine Particulate Matter (PM_{2.5}), the Exposome, and Cumulative Health Risk: A Life-course Perspective

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Fine particulate matter (PM_{2.5}; aerodynamic diameter $\leq 2.5 \mu\text{m}$) is a leading contributor to the global burden of disease and is associated with substantial morbidity and mortality. However, its health effects are often evaluated within single-exposure frameworks, limiting causal inference and reducing the translational relevance of evidence for policy development. In this study, we reconceptualise PM_{2.5} within the exposome framework by integrating external exposures, internal biological responses, and structural determinants of health across the life course. PM_{2.5} comprises a complex and dynamic mixture of pollutants from ambient and household sources, including biomass combustion, traffic emissions, and industrial activities. After inhalation, PM_{2.5} particles can penetrate the respiratory tract, and finer fractions may enter the systemic circulation, where they induce oxidative stress, inflammation, epigenetic dysregulation, and endocrine disruption. These mechanisms contribute to a broad spectrum of adverse health outcomes, including cardiovascular, respiratory, metabolic, and neurodegenerative diseases as well as cancers. The multi-level and temporally accumulated nature of these effects highlights the limitations of reductionist exposure assessment and underscores the need for integrative, systems-based approaches. Integrating cumulative impact assessment with exposome-wide approaches that leverage satellite, clinical, land-use, urban planning, and socioeconomic data can improve the characterization of context-specific cumulative health effects. Positioning PM_{2.5} as a sentinel indicator of cumulative environmental risk provides a unifying framework for exposome-informed science and supports the development of more precise, equitable, and policy-relevant interventions.

Key words: Fine particulate matter (PM_{2.5}); Exposome; Cumulative risk; Toxic mixtures; Life course; Environmental health

INTRODUCTION

Fine particulate matter (PM_{2.5}; aerodynamic diameter $\leq 2.5 \mu\text{m}$) remains a major global environmental health concern, yet much

of the literature continues to examine its effects as isolated exposure–outcome relationships. PM_{2.5} is a complex mixture comprising gaseous pollutants, such as ozone (O₃), sulphur oxides (SO_x), and nitrogen oxides (NO_x); heavy metals; organic compounds, including volatile organic compounds and polycyclic aromatic hydrocarbons; and emerging contaminants, including microplastics and perfluoroalkyl and polyfluoroalkyl substances. This complexity challenges pollutant-specific approaches, which may preserve methodological rigour but incompletely capture the broader exposure context. Such approaches also align with existing regulatory frameworks, although they rarely interrogate the assumptions underlying

Received: Mar 10, 2026 Revised: Apr 27, 2026 Accepted: May 19, 2026

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those frameworks. Consequently, the resulting evidence may appear robust while remaining limited in its capacity to guide interventions or identify critical windows of vulnerability.

Positioning PM_{2.5} within the exposome framework, which integrates multiple exposures, biological responses, and social determinants, preserves scientific rigour and policy relevance while better reflecting its role as a sentinel indicator of cumulative environmental exposure across the lifespan. Long-term PM_{2.5} exposure is associated with cardiovascular and respiratory diseases, lung cancer, metabolic disorders, and neurodegeneration. Early-life exposure contributes to adverse birth outcomes and childhood asthma, whereas cumulative adult exposure is associated with an increased risk of cardiovascular disease, chronic obstructive pulmonary disease, and cognitive decline. Globally, combined ambient and household exposure to PM_{2.5} was estimated to account for more than 4 million deaths and more than 100 million disability-adjusted life-years in 2017 [1]. Moving beyond pollutant-specific paradigms requires examining PM_{2.5} within the exposome framework, which captures cumulative exposures, biological embedding, and structural determinants across the life course. As a sentinel marker of interacting environmental, social, and biological processes, PM_{2.5} reveals complex pathways that traditional approaches often overlook, particularly in heavily exposed low-income and middle-income regions. This perspective strengthens understanding of cumulative risk, supports earlier detection and targeted prevention, and informs more effective public health policies. By enabling dynamic and individualised risk assessment, the exposome framework also fosters interdisciplinary research and facilitates the development of improved tools for disease prevention [2].

FINE PARTICULATE MATTER IN CUMULATIVE RISK ASSESSMENT: AN INTEGRATIVE FRAMEWORK

PM_{2.5} is both a major air pollutant and a key component of the exposome. This pollutant reflects combined exposure to environmental stressors arising from biomass burning, transportation, indoor cooking, fossil fuel combustion, and industrial emissions, especially in low-income settings. Their small size allows PM_{2.5} particles to penetrate deep into the alveoli and enter the systemic circulation, where they affect multiple organ systems [3]. PM_{2.5} serves as a sentinel indicator of cumulative environmental risk by integrating exposures from diverse

combustion sources into a single, routinely monitored metric. Its spatiotemporal patterns track population exposure to complex pollutant mixtures; consistent associations with multiple health outcomes further support its use as a sensitive, policy-relevant indicator of underlying cumulative risk. Interactions among co-pollutants can amplify biological responses beyond those associated with PM_{2.5} mass alone. Mechanistically, these exposures promote oxidative stress, inflammation, epigenetic dysregulation, mitochondrial dysfunction, and endocrine disruption; some components may also bioaccumulate [4]. These interactions may explain spatial differences in health outcomes, even at similar PM_{2.5} concentrations. Long-term exposure is also associated with key exposome pathways, including systemic inflammation, endothelial dysfunction, autonomic imbalance, and epigenetic modification [5]. Recent research has associated PM_{2.5} exposure with changes in DNA methylation and microRNA expression, which may contribute to long-term disease risk even decades after the initial exposure [5]. Technological advances—including satellite remote sensing, chemical transport models, personal sensors, and land-use regression—have improved the characterisation of PM_{2.5} exposure and enabled high-resolution spatial and temporal estimates. These tools support more comprehensive exposure assessment and facilitate the integration of PM_{2.5} into exposome-wide association studies, thereby strengthening evidence on health impacts [6]. However, the community-level health effects of PM_{2.5} remain incompletely understood because conventional risk assessments focused on single endpoints often fail to capture its interactions with other environmental stressors and socioeconomic vulnerabilities.

Cumulative impact assessment (CIA) integrates multisource data, including satellite observations, clinical records, socioeconomic indicators, and community knowledge, to identify context-specific risks. Coupled with big data analytics, CIA can enable scalable, more equitable risk assessment and inform evidence-based policies to reduce exposure and advance environmental justice. However, its application remains constrained by fragmented data systems, limited incorporation of community knowledge, and the absence of standardised tools. These limitations highlight the need for stronger integration into policy and regulatory frameworks to generate actionable insights that support equitable and sustainable community well-being [7].

LIFESPAN HEALTH IMPACTS FROM TOXIC MIXTURES OF FINE PARTICULATE MATTER EXPOSURE: AN EXPOSOME-BASED PERSPECTIVE

The exposome encompasses the full range of environmental and lifestyle exposures across the life course. By contrast, cumulative impacts refer to the aggregate health effects of concurrent exposure to multiple chemical and non-chemical stressors and to their combined influence on health, well-being, and

quality of life [8]. PM_{2.5} exposure is associated with a continuum of adverse health effects across the life course, from foetal development to ageing. The exposome framework captures this temporal and biological complexity and highlights critical windows of susceptibility and cumulative risk. Exposure in early life is associated with adverse birth outcomes and impaired development, whereas long-term exposure contributes to cardiovascular, respiratory, and metabolic diseases along with cancers. Exposure disparities are well documented and remain substantial; the highest burdens are observed in low-income

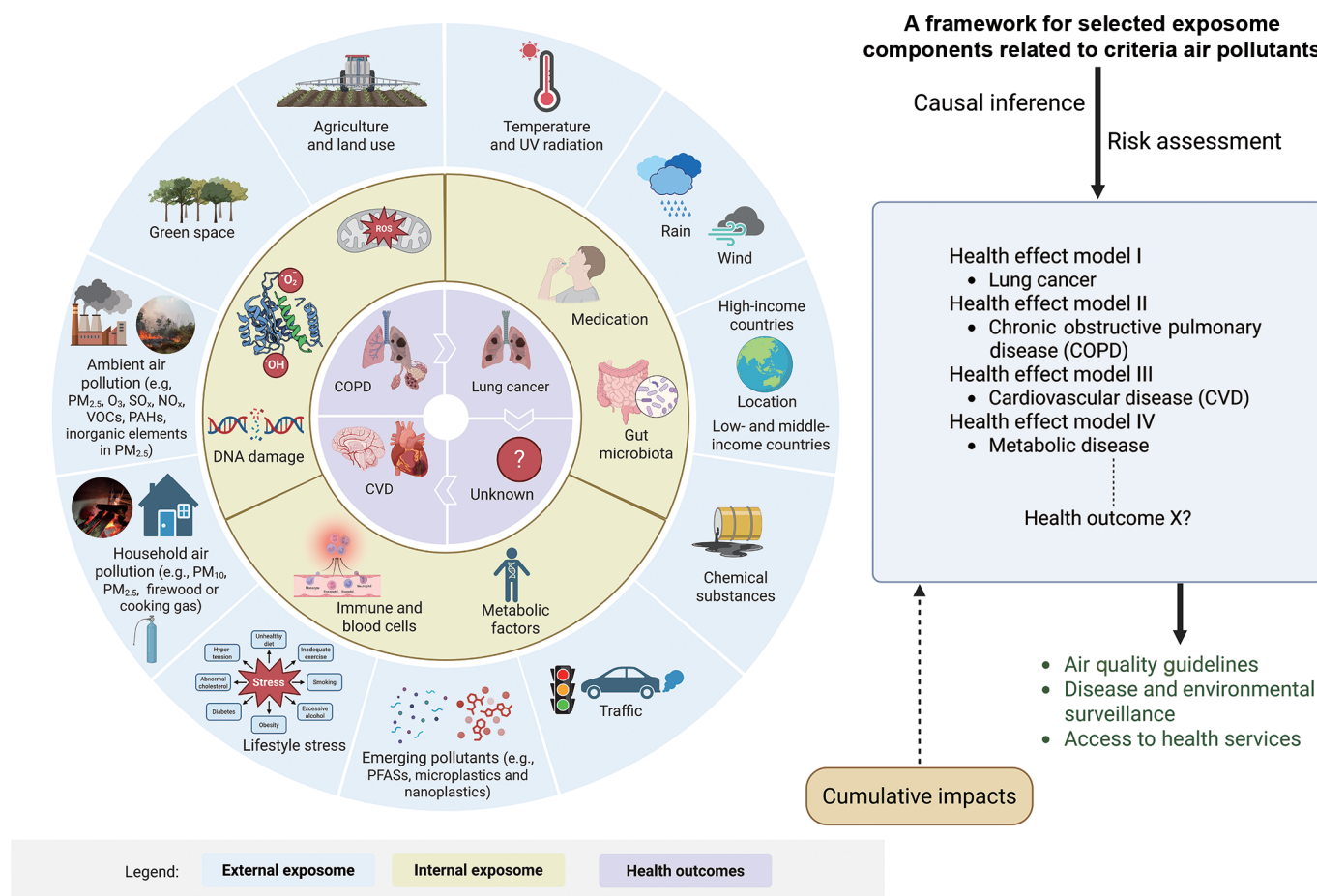


Figure 1. Conceptual exposome framework for long-term PM_{2.5} exposure and cumulative health risk across the life course. PM_{2.5} is positioned as a sentinel exposure arising from multiple external sources and co-exposures, including ambient and household air pollution, chemical contaminants, climate-related factors, and social or geographic context. External exposures (the “external exposome”), including PM_{2.5}, volatile organic compounds (VOCs), polycyclic aromatic hydrocarbons (PAHs), and perfluoroalkyl and polyfluoroalkyl substances (PFASs), interact with internal biological processes (the “internal exposome”), including the generation of reactive oxygen species (ROS), such as superoxide (O₂⁻) and hydroxyl radicals (•OH), as well as oxidative stress, inflammation, immune dysregulation, DNA damage, gut microbiome alterations, and metabolic dysfunction. These pathways influence the risk of non-communicable diseases, including CVD, COPD, lung cancer, and other chronic diseases. PM_{2.5}, fine particulate matter with an aerodynamic diameter of ≤2.5 μm; PM₁₀, particulate matter with an aerodynamic diameter of 10 μm or less; O₃, ozone; SO_x, sulphur oxides; NO_x, nitrogen oxides; UV, ultraviolet. Adapted from American Lung Association. Cumulative health impacts assessment and the national ambient air quality standards; 2024 [Internet] [10]. Created with BioRender.com.

and middle-income regions and disproportionately affect women and children. For example, Nobile et al. [9] demonstrated that combined exposures—higher air pollution and lower greenness—were associated with increased mortality across European cohorts and that cumulative exposome risk amplified these effects. These findings underscore the importance of addressing interacting environmental exposures to improve population health. However, important evidence gaps persist, particularly regarding diverse co-exposures and underrepresented populations. Closing these gaps will require strengthened research capacity, expanded air-quality monitoring, and policies that prioritise public health at local, national, and global scales. Accordingly, we position PM_{2.5} as a central exposure within an exposome-based cumulative risk paradigm. Within this framework, we distinguish (1) external exposures, including PM_{2.5} mixtures and co-pollutants; (2) the internal exposome, including oxidative stress, inflammation, and epigenetic alterations; and (3) health outcomes across the life course. Socioeconomic and structural determinants are conceptualized as upstream drivers that shape exposure and vulnerability and thereby influence cumulative impacts. We further present a conceptual model that integrates key exposome components related to criteria air pollutants, adapted from the American Lung Association (Figure 1) [10].

FUTURE DIRECTIONS

Future exposome research should include relevant co-pollutants such as ultrafine particulate matter (PM_{0.1}), O₃, SO₂, and NO₂ along with PM_{2.5} to better capture the real-world multi-pollutant exposures. There is a need for more emphasis on understanding the interactions between external exposures and internal biological responses and on developing integrated, multiscale exposure models. Given that the majority of exposome platforms continue to be centered around high-income settings, research should also be expanded to settings with more complex or larger pollution mixtures. Key priorities are real-time and longitudinal exposure tracking and ethically sound approaches to personal and digital monitoring.

CONCLUSION

Reframing PM_{2.5} within the exposome offers a more realistic public health model for understanding cumulative environmental risk. Rather than treating PM_{2.5} as an isolated pollutant,

this perspective recognises it as part of a dynamic network of co-exposures, biological responses, and social conditions that shape disease risk across the life course. Progress in this field will require harmonised exposure data, longitudinal cohort designs, biomarkers that indicate susceptibility and response, and analytic frameworks capable of integrating cumulative impacts across populations. Together, these efforts can strengthen prevention strategies, refine environmental risk assessment, and inform policies to reduce health inequities associated with short- and long-term exposure to air pollution.

Ethics Statement

This paper is a perspective article; therefore, ethical approval was not required.

NOTES

Conflict of Interest

The authors have no conflicts of interest associated with the material presented in this paper.

Funding

None.

Acknowledgements

None.

Author Contributions

Conceptualization: Sangkham S. Formal analysis: Sangkham S. Funding acquisition: None. Methodology: Sangkham S. Visualization: Sangkham S. Writing – original draft: Sangkham S. Writing – review & editing: Sangkham S, Lonan P, Munkong N, Thim-Uam A, Phairuang W.

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